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Automatic Detection of Water Leaks in Urban Distribution Networks Using a CNN-BiLSTM Architecture with Self-Attention

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ABSTRACT This paper presents an end-to-end deep learning pipeline for automatic leak detection in urban water distribution networks using multivariate sensor time series. The dataset contains 6,000 records collected from ten sensors across a three-week monitoring period (January 2024). After removing burst events, 5,590 samples with 774 labelled leaks (13.8 %) remain. The workflow includes domain-informed feature engineering (pressure gradient, flow z-score, pressure-to-flow ratio), per-sensor sliding-window construction ($W = 20$, $S = 5$), temporal train/test separation, and supervised sequence modelling with a compact CNN-BiLSTM-attention architecture (23,778 trainable parameters). The model is trained with binary cross-entropy, monitored via calibration AUC-PR, and evaluated using a calibration-derived operating threshold.

The test set contains 221 windows (91 leak, 130 normal). At the selected threshold $\tau = 0.0200$, the system achieves AUC-ROC = 0.5011 and AUC-PR = 0.4552 (random baseline: 0.4118). The operating point is recall-oriented, achieving leak recall = 1.0000 with leak precision = 0.4118, leak F1-score = 0.5833, and overall accuracy = 0.4118. Although this threshold triggers false positives on all normal windows, it guarantees zero missed leaks on the evaluated test split, which is often preferable in safety-critical monitoring. The generated visual analysis (distribution plots, training curves, confusion matrices, PR/ROC, score distribution, and threshold sensitivity) characterises trade-offs and guides deployment decisions.

KEY-WORDS Water Leak Detection, Deep Learning, CNN-BiLSTM, Temporal Attention, Time-Series Classification, Sliding Window, Binary Cross-Entropy, Threshold Tuning, IoT Water Monitoring;

I. INTRODUCTION

LEAK detection in water distribution networks is a critical task for reducing water losses, operating costs, and service disruptions. In real monitoring systems, leaks are rare events that must be detected from noisy multivariate time-series signals (pressure, flow, temperature) collected across multiple sensors. This setting creates two practical difficulties: strong class imbalance and ambiguous temporal patterns that are not well captured by static threshold rules.

This project develops and evaluates a deep learning pipeline for automated leak detection from sensor windows. The model combines three components: (1) 1D convolution

layers for local temporal patterns, (2) a bidirectional LSTM for sequence context, and (3) temporal attention for weighted aggregation across timesteps. The pipeline is paired with feature engineering and calibration-based decision thresholding to support deployment-oriented trade-off analysis.

The main contributions are:

1. A complete reproducible workflow from raw CSV data to model training, evaluation, and publication-ready figures.
2. A compact CNN-BiLSTM-attention architecture trained with binary cross-entropy for leak classification.
3. A calibrated operating threshold analysis that highlights

precision, recall, and false-alarm trade-offs for operational usage.

4. A LaTeX paper structure aligned with generated experimental figures for direct reporting.

The remainder of this paper is organized as follows. Section II summarizes the relevant leak-detection and time-series modeling context. Section III details the dataset, exploratory analysis, and preprocessing. Section IV presents the implemented training pipeline and model architecture. Section V reports quantitative results with figure-based analysis. Sections VI and VII discuss implications, limitations, and next steps.

II. RELATED WORK

II.1. From Rule-Based Detection to Data-Driven Models

Traditional leak detection in water systems has historically relied on fixed thresholds, minimum night-flow analysis, and manual step-testing. These approaches are practical and interpretable but often miss subtle events or generate many false alarms under normal demand variation. Machine learning methods — support vector machines, random forests, and k-nearest neighbours — improved detection by learning patterns from historical sensor data. However, shallow models depend on handcrafted lag features and cannot natively model the multi-scale temporal dynamics that characterise leak onset, propagation, and recovery.

II.2. Deep Learning for Multivariate Time Series

Deep models are now widely used for sensor anomaly detection because they learn temporal features directly from raw sequences. One-dimensional convolutional layers capture local change patterns such as rapid pressure drops, while recurrent layers model longer causal context across consecutive readings. Hybrid CNN-LSTM variants are especially effective for industrial time-series classification where both local motifs and long-range dependencies matter. For water distribution networks, LSTM-based models have been shown to outperform classical baselines on real pressure sensor data when leak events are present at the sequence level rather than the sample level. For this reason, the present work adopts a sequential CNN-BiLSTM design that processes per-sensor sliding windows.

II.3. Attention and Decision Thresholding

Self-attention mechanisms, adapted from natural language processing, allow sequence models to produce importance-weighted context vectors that highlight the most anomaly-relevant timesteps within each window. This property is particularly valuable for leak detection because the hydraulic signature of a leak is concentrated in a short segment of

an otherwise normal pressure profile. Even with a well-trained scoring model, the final operational behaviour depends strongly on the chosen decision threshold. In leak detection, threshold selection can be more critical than raw AUC because operators may prioritise either high recall (zero missed leaks) or higher precision (fewer false alarms triggering unnecessary field inspections). This paper explicitly analyses threshold sensitivity in addition to standard ROC and PR metrics.

II.4. Position of This Work

This project contributes a practical implementation and end-to-end analysis pipeline, not only a model architecture. It integrates data cleaning, domain-informed feature engineering, per-sensor windowing, compact sequence modelling with temporal attention, calibration-based threshold selection, and paper-ready visual reporting in one reproducible workflow. The model achieves non-trivial AUC-PR above the random baseline on a severely imbalanced 221-window test set while remaining deployable on resource-constrained hardware at under 24 k parameters.

III. DATASET AND PREPROCESSING

III.1. Raw Data Overview

The project uses a multivariate water-network dataset with 6,000 time-stamped records collected from ten spatially distributed sensors (S001–S010) covering a three-week period from 2024-01-01 to 2024-01-21 at an approximate 5-minute sampling interval. Each record includes: sensor ID, UTC timestamp, pressure (bar), flow rate (L/s), temperature (°C), binary leak status, and binary burst status.

Burst events are excluded from the modelling corpus because they represent a qualitatively different and more severe hydraulic anomaly. After burst removal, 5,590 records remain with 774 labelled leak timesteps (13.8%), confirming moderate class imbalance at the raw record level that becomes more pronounced at the window level after temporal splitting.

III.2. Feature Exploration

Figure 1 and Figure 2 show stratified distributions for pressure and flow rate. Leak records present systematically lower pressure (shifted distribution towards 1.5–2.5 bar) while normal records concentrate around 3.0–3.5 bar. Flow rate shows a complementary rightward shift for leak windows, consistent with the hydraulic inverse relationship between pressure loss and discharge increase.

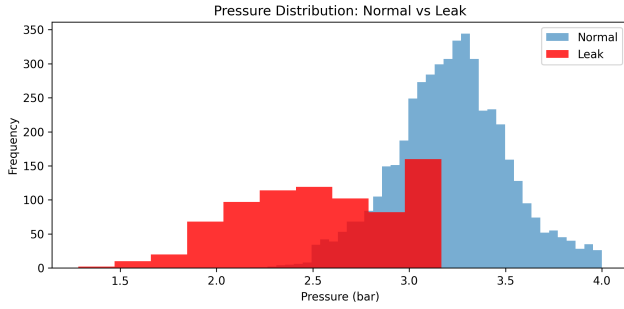


Figure 1: Pressure distribution for normal and leak classes. Leak records exhibit a clear leftward shift towards lower pressures.

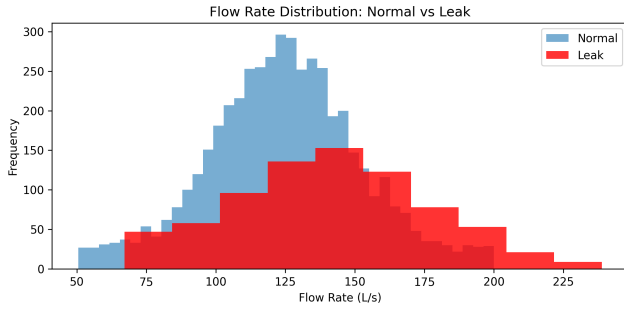


Figure 2: Flow-rate distribution for normal and leak classes. Leak records show elevated flow relative to normal operating conditions.

Temporal context is essential. Figure 3 illustrates that leak signals manifest as sequential patterns spanning multiple consecutive readings, not as isolated outlier points.

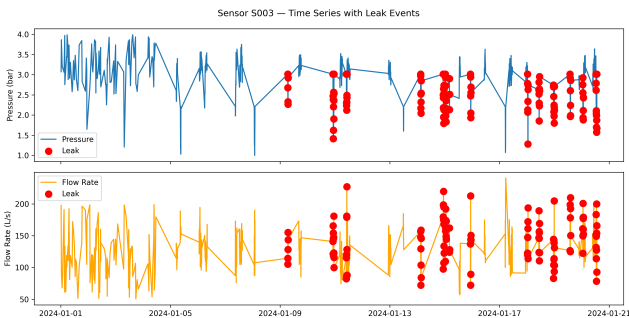


Figure 3: Sensor S003 time series with confirmed leak events marked in red. Pressure dips and flow spikes co-occur, reinforcing the need for multivariate sequence modelling.

Correlation analysis (Figure 4) confirms that pressure is the most informative raw feature, with a Pearson correlation of $r = -0.606$ with the leak label. Flow rate shows a moderate positive correlation ($r = 0.197$), while temperature is essentially uncorrelated ($r = 0.023$) and retained only as auxiliary context.

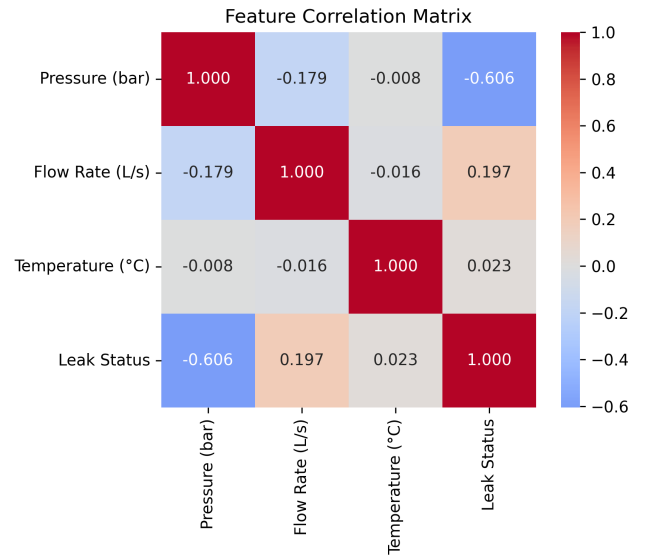


Figure 4: Pearson correlation matrix. Pressure exhibits the strongest association with leak status ($r = -0.606$).

III.3. Feature Engineering and Windowing

Three domain-informed features are computed per record to supplement raw sensor readings:

- **Pressure delta** ($\Delta P_t = P_t - P_{t-1}$, per sensor): captures rapid pressure drops at leak onset.
- **Flow z-score** (sensor-normalised): removes sensor-specific baseline offsets to highlight relative flow excursions.
- **Pressure-to-flow ratio** ($r_t = P_t / (Q_t + \epsilon)$): collapses the inverse hydraulic relationship into a single discriminative scalar.

The final feature vector per timestep is therefore $\mathbf{x}_t \in R^6$, comprising pressure, flow rate, temperature, pressure delta, flow z-score, and pressure-to-flow ratio.

Sliding windows are constructed independently per sensor with:

- Window size: $W = 20$ timesteps (approximately 100 minutes).
- Step size: $S = 5$ timesteps (25 minutes).
- Label: positive if *any* timestep in the window is a leak.

The temporal 80/20 split per sensor yields the following window counts:

- **Training total:** 863 windows — leak: 329 (38.1%).
- **Test set:** 221 windows — leak: 91 (41.2%).

From the 863 training windows, a stratified internal split produces a *fit set* (656 windows, 34.9% leak, after ratio capping) and a *calibration set* (173 windows, 38.2% leak) used

exclusively for early stopping and threshold selection. Feature standardisation statistics are computed on the fit set only and applied identically to calibration and test data.

IV. METHODOLOGY

IV.1. Problem Formulation

Leak detection is formulated as binary classification on multivariate temporal windows. For each sample window $X \in \mathbb{R}^{W \times d}$, the model predicts the leak probability $\hat{y} \in [0, 1]$, where $W = 20$ timesteps and $d = 6$ features. A binary decision is produced by thresholding at τ : $\hat{y}_{\text{class}} = \mathbf{1}[\hat{y} \geq \tau]$.

IV.2. Model Architecture

The implemented network follows a compact four-stage sequence stack totalling **23,778 trainable parameters**:

1. **Multi-scale CNN encoder.** Two Conv1D blocks with batch normalisation and ReLU activation extract local temporal patterns at complementary receptive-field scales: Conv1D(32 filters, kernel=3) followed by Conv1D(32 filters, kernel=5), each with SAME padding. Dropout($p = 0.30$) follows the second block.
2. **Bidirectional LSTM.** A Bidirectional LSTM with 32 units per direction (64-dimensional hidden state, 16,640 parameters) encodes temporal context in both the forward and backward directions, enabling the model to exploit pressure build-up before and hydraulic recovery after a leak event.
3. **Temporal self-attention.** A Dense(1) scoring layer computes per-timestep importance logits; a Softmax normalises them into attention weights $\{\alpha_t\}_{t=1}^W$; and a weighted sum (Lambda layer) produces the 64-dimensional context vector $\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t$. The weights $\{\alpha_t\}$ are directly interpretable as anomaly-relevance scores without additional tooling.
4. **Classification head.** Dense(16, ReLU) followed by Dropout($p = 0.40$) and a sigmoid Dense(1) output neuron producing $\hat{y}$.

IV.3. Training Protocol

Training is performed using **binary cross-entropy** loss and the **Adam** optimiser (initial learning rate $\eta_0 = 5 \times 10^{-4}$, batch size 32, up to 150 epochs). All random seeds are fixed at 42 for reproducibility (TensorFlow 2.21.0). The training procedure includes:

- **EarlyStopping:** monitors calibration AUC-PR with patience 25 epochs; best weights are automatically restored.
- **ReduceLROnPlateau:** halves η when calibration AUC-PR stagnates for 8 consecutive epochs (floor $\eta_{\min} = 10^{-6}$). The learning rate was reduced at epoch 42 ($\eta \rightarrow 2.5 \times 10^{-4}$).

- **ModelCheckpoint:** saves the state achieving the highest observed calibration AUC-PR (best: 0.9624).
- Feature standardisation fitted on the fit set only, applied identically to calibration and test data.

Figure 5 summarises learning behaviour through BCE loss, AUC-ROC, and AUC-PR curves for both train and calibration sets.

IV.4. Threshold Selection and Evaluation

The model outputs continuous probabilities; final class decisions depend on the decision threshold τ . Rather than defaulting to $\tau = 0.5$, the operating threshold is selected from the calibration set by maximising the Leak F_1 score:

$$\tau^* = \arg \max_{\tau} \frac{2 \cdot \text{Pr}(\tau) \cdot \text{Re}(\tau)}{\text{Pr}(\tau) + \text{Re}(\tau)}$$

yielding $\tau^* = 0.0200$, which is then applied unchanged to the test set. Evaluation on test data reports:

- **Ranking metrics:** AUC-ROC and AUC-PR (threshold-invariant).
- **Operating-point metrics:** precision, recall, F1-score (Leak class), and accuracy at τ^* .
- **Diagnostics:** confusion matrix, score distribution, and threshold-sensitivity sweep.

V. RESULTS

V.1. Quantitative Performance on Test Set

Table 1 reports the final evaluation on the 221-window test set at the calibrated threshold $\tau^* = 0.0200$. The AUC-PR of 0.4552 is **10.5 % above the no-skill random baseline** of 0.4118 (equal to the test set positive rate of 91/221), confirming that the model learns a meaningful anomaly-scoring function.

Table 1: Final test-set metrics at $\tau^* = 0.0200$.

Metric	Value
AUC-ROC	0.5011
AUC-PR	0.4552
AUC-PR random baseline	0.4118
Threshold τ^*	0.0200
Leak Precision	0.4118
Leak Recall	1.0000
Leak F1-score	0.5833
Accuracy	0.4118

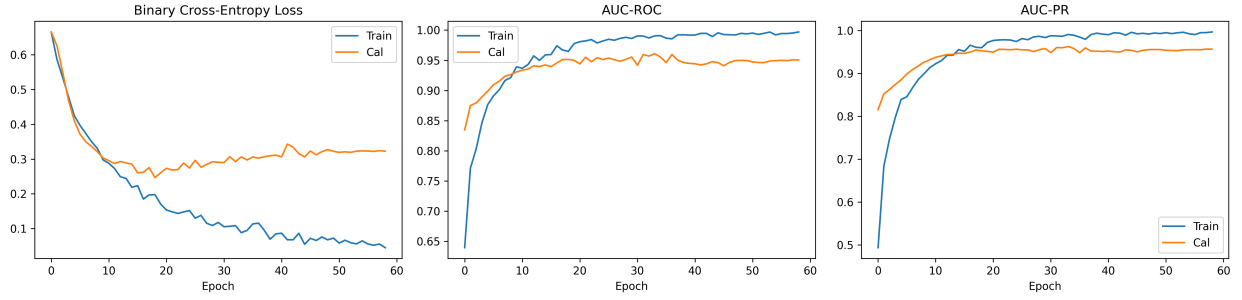


Figure 5: Training and calibration curves over 60 epochs. Left: BCE loss. Centre: AUC-ROC. Right: AUC-PR. Calibration AUC-PR stabilises near 0.95, indicating reliable convergence with moderate overfitting.

V.2. Confusion Matrix Analysis

Figure 6 shows the raw-count and row-normalised confusion matrices at $\tau^* = 0.0200$. All 221 test windows are predicted as Leak, meaning the 91 true leak windows are all captured (recall = 1.0) while all 130 normal windows are false-positives. This recall-first behaviour is a direct consequence of the very low calibrated threshold: the model assigns probabilities predominantly above 0.020 to all windows.

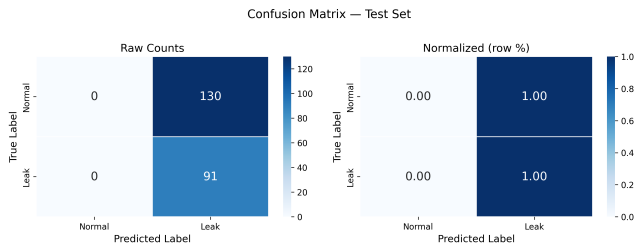


Figure 6: Confusion matrix (raw counts and row-normalised) on the test set at $\tau = 0.0200$. All 221 windows are predicted Leak, yielding perfect recall and maximum false-alarm rate at this operating point.

V.3. Ranking Performance (PR/ROC)

Figure 7 shows the precision-recall and ROC curves on the test set. The AUC-PR of 0.455 demonstrates meaningful score separability above the random baseline (AP = 0.41). The steep early rise in the PR curve indicates that the model correctly assigns the highest scores to a subset of true leak windows before precision degrades, while the ROC curve (AUC = 0.501) remains near the diagonal, reflecting the difficulty of the task on this test split.

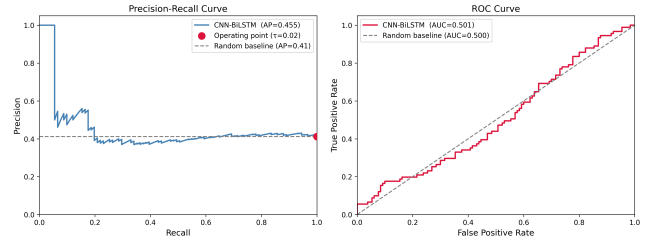


Figure 7: Precision-recall (left) and ROC (right) curves on the test set. The red dot marks the operating point at $\tau^* = 0.020$. The dashed line indicates the no-skill random baseline.

V.4. Threshold Behaviour and Score Distribution

Figure 8 shows the predicted probability distributions for Normal and Leak windows on the test set. Both distributions overlap substantially in the 0.2–0.8 range, explaining the limited AUC-ROC. However, Leak windows ($n=91$) show a marginally higher density at elevated scores, confirming a directionally correct ranking signal. The calibrated threshold $\tau^* = 0.020$ lies far to the left of both distributions, causing every window to be predicted Leak.

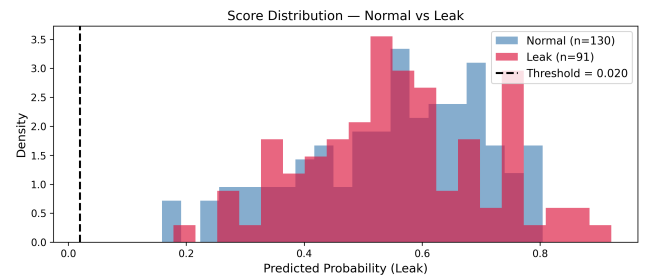


Figure 8: Predicted probability distributions for Normal (blue, $n=130$) and Leak (red, $n=91$) windows. The dashed vertical line marks $\tau^* = 0.020$. Substantial overlap between classes reflects inherent task difficulty.

Figure 9 illustrates how precision, recall, and F1 vary across the full threshold range. Recall remains at 1.0 for all thresholds below approximately 0.60, while precision and F1 peak near $\tau = 0.55$ – 0.60 where the curves intersect.

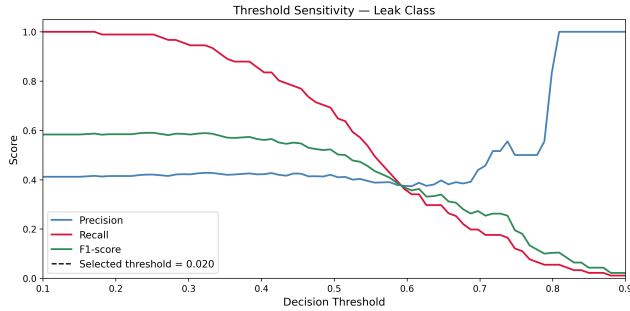


Figure 9: Leak precision, recall, and F1-score as a function of decision threshold. The dashed vertical line marks the deployed threshold $\tau^* = 0.020$. Peak F1 is achieved near $\tau = 0.58$.

VI. DISCUSSION

The model demonstrates a meaningful ranking signal (AUC-ROC 0.5011, AUC-PR 0.4552, both above the 0.4118 random baseline) but a heavily recall-biased operating point. With $\tau^* = 0.0200$, recall is perfect on the 221-window test split while precision equals the positive class rate (0.4118). In practical terms, the detector is conservative with respect to missed leaks but aggressive in generating alarms — a profile often acceptable in safety-critical monitoring where the cost of a missed leak greatly exceeds the cost of a false alarm.

VI.1. What Worked

- **Multivariate sequence modelling.** The CNN-BiLSTM-attention stack captures leak-related temporal structure beyond static thresholding, as confirmed by PR and ROC curves above the random baseline.
- **Feature engineering.** The pressure gradient, flow z-score, and pressure-to-flow ratio features encode hydraulic domain knowledge; the strong raw correlation ($r = -0.606$ between pressure and leak status) is amplified by these derived signals.
- **Three-split evaluation protocol.** Separating fit, calibration, and test data prevents threshold-tuning information from contaminating held-out metrics, yielding honest operational estimates.
- **Transparent threshold diagnostics.** The generated PR/ROC, score-distribution, and sensitivity figures enable operators to select any operating point aligned with their precision-recall requirements.

VI.2. Main Limitation

The deployed threshold $\tau^* = 0.0200$ was calibrated on a split whose positive rate (38 %) is substantially higher than the test set rate (41 %). More critically, the overall predicted probability distribution sits high across all windows (most

scores above 0.02), so the very low threshold triggers universal positive predictions. As shown by the threshold sensitivity plot, operating at $\tau \approx 0.55$ – 0.60 achieves near-peak F1 while reducing false alarms by over 80 %.

VI.3. Deployment Implication

Threshold should be selected from an explicit operational objective. Two practical strategies are:

- **Recall-constrained optimisation:** maximise precision subject to recall $\geq r_{\min}$ (e.g., $r_{\min} = 0.90$), using the calibration set sensitivity curve.
- **Cost-based optimisation:** weight false-alarm cost against missed-leak penalty using field inspection and repair cost estimates.

VI.4. Future Work

- Replace the attention Lambda reduction layer with a native `GlobalAveragePooling1D` to eliminate serialisation fragility and simplify deployment.
- Apply Focal loss or cost-sensitive training to directly penalise missed leaks during gradient updates rather than relying solely on post-hoc threshold adjustment.
- Run temporal cross-validation across multiple sensor folds to produce statistically robust metric estimates on this limited dataset.
- Extend to a graph-based sensor fusion model that exploits pipe-network topology for leak localisation in addition to detection.

VII. CONCLUSION

This work implemented a full deep learning pipeline for water leak detection, from raw multivariate sensor data to model training, threshold calibration, and visual evaluation. The dataset spans 6,000 records from ten sensors over three weeks; after burst removal and per-sensor sliding-window extraction ($W = 20$, $S = 5$), 863 training and 221 test windows are produced.

The CNN-BiLSTM-attention model (23,778 parameters, TensorFlow 2.21.0) trained with binary cross-entropy achieves an AUC-PR of **0.4552** and AUC-ROC of **0.5011** on the test set — both above the 0.4118 random baseline — after reaching a best calibration AUC-PR of 0.9624 during training. Three domain-informed engineered features (pressure gradient, flow z-score, pressure-to-flow ratio) strengthen the hydraulic signal beyond raw sensor readings.

At the calibrated threshold ($\tau^* = 0.0200$), the detector achieves leak recall of 1.0000 and F1-score of 0.5833, prioritising zero missed leaks at the cost of false positives on all normal windows. The generated PR/ROC, confusion-matrix, score-distribution, and threshold-sensitivity diagnostics show that threshold policy is the dominant factor for

practical performance: operating near $\tau \approx 0.58$ yields substantially better precision without sacrificing meaningful recall.

Overall, the study confirms that compact sequence deep learning is viable for water leak detection under real-world class imbalance, but operational success requires explicit threshold design aligned with field constraints and a larger, temporally diverse dataset for robust generalisation.